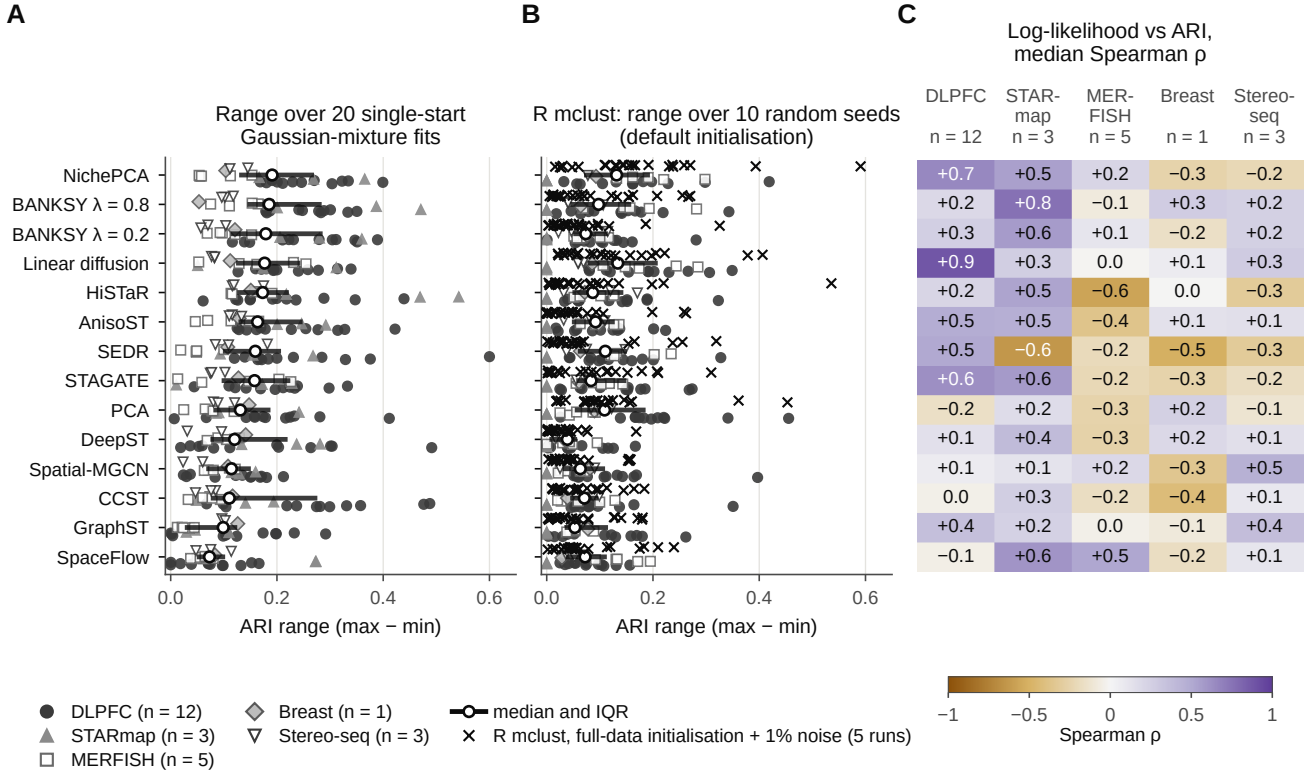

A
Range over 20 single-start Gaussian-mixture fits
NichePCA
BANKSY λ = 0.8
BANKSY λ = 0.2
Linear diffusion
HiSTaR
AnisoST
SEDR
STAGATE
PCA
DeepST
Spatial-MGCN
CCST
GraphST
SpaceFlow
0.0
0.2
0.4
0.6
ARI range (max − min)
B
R mclust: range over 10 random seeds (default initialisation)
0.0
0.2
0.4
0.6
ARI range (max − min)
C
Log-likelihood vs ARI, median Spearman ρ
DLPFC n = 12
STAR-map n = 3
MER-FISH n = 5
Breast n = 1
Stereo-seq n = 3
+0.7 +0.5 +0.2 −0.3 −0.2
+0.2 +0.8 −0.1 +0.3 +0.2
+0.3 +0.6 +0.1 −0.2 +0.2
+0.9 +0.3 0.0 +0.1 +0.3
+0.2 +0.5 −0.6 0.0 −0.3
+0.5 +0.5 −0.4 +0.1 +0.1
+0.5 −0.6 −0.2 −0.5 −0.3
+0.6 +0.6 −0.2 −0.3 −0.2
−0.2 +0.2 −0.3 +0.2 −0.1
+0.1 +0.4 −0.3 +0.2 +0.1
+0.1 +0.1 +0.2 −0.3 +0.5
0.0 +0.3 −0.2 −0.4 +0.1
+0.4 +0.2 0.0 −0.1 +0.4
−0.1 +0.6 +0.5 −0.2 +0.1
DLPFC (n = 12)
STARmap (n = 3)
MERFISH (n = 5)
Breast (n = 1)
Stereo-seq (n = 3)
median and IQR
R mclust, full-data initialisation + 1% noise (5 runs)
−1
−0.5
0
0.5
1
Spearman ρ